

Toward Scalable and Valid Conditional Independence Testing with Spectral Features

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Problem

We all know there is no free lunch in conditional independence testing

$$\mathcal{H}_0 : X \perp\!\!\!\perp Y \mid Z \quad \text{vs.} \quad \mathcal{H}_1 : X \not\perp\!\!\!\perp Y \mid Z$$

SoTA is method-centric, with limited real-world data and theoretical guarantees

Approach

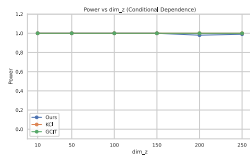
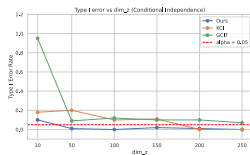
By mixing spectral contrastive learning with partial covariance operators

$$\Sigma_{X\ddot{Y}.Z} = \Sigma_{X\ddot{Y}} - \Sigma_{XZ}\Sigma_{Z\ddot{Y}},$$

under null, $\hat{T}_n = n\|\hat{\Sigma}_{X\ddot{Y}.Z}\|_F^2 \xrightarrow{d} \chi^2$; under alternative, power from repr learning

Roadmap

- ▶ Extending theory to finite-samples
- ▶ Working with genetics and image data



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